

Quarterly Business Review

What Drives Us

Excellence

High standards in every deliverable we ship.

Integrity

Transparent and honest in all we do.

Innovation

Investing in ideas that shape tomorrow.

By the Numbers

127%

Revenue Target

94%

Retention Rate

48 NPS

Net Promoter

Side-by-Side Evaluation

Legacy Platform

Deploy time: 4 hours

Manual scaling

On-premise; high cost

Modern Stack

Deploy time: 8 min

Auto-scales fast

Cloud-native; low cost

Key Milestones

Q1

Foundation

Cloud platform migration done.

Q2

API Launch

REST API live with 99.99% SLA.

Q3

AI Beta

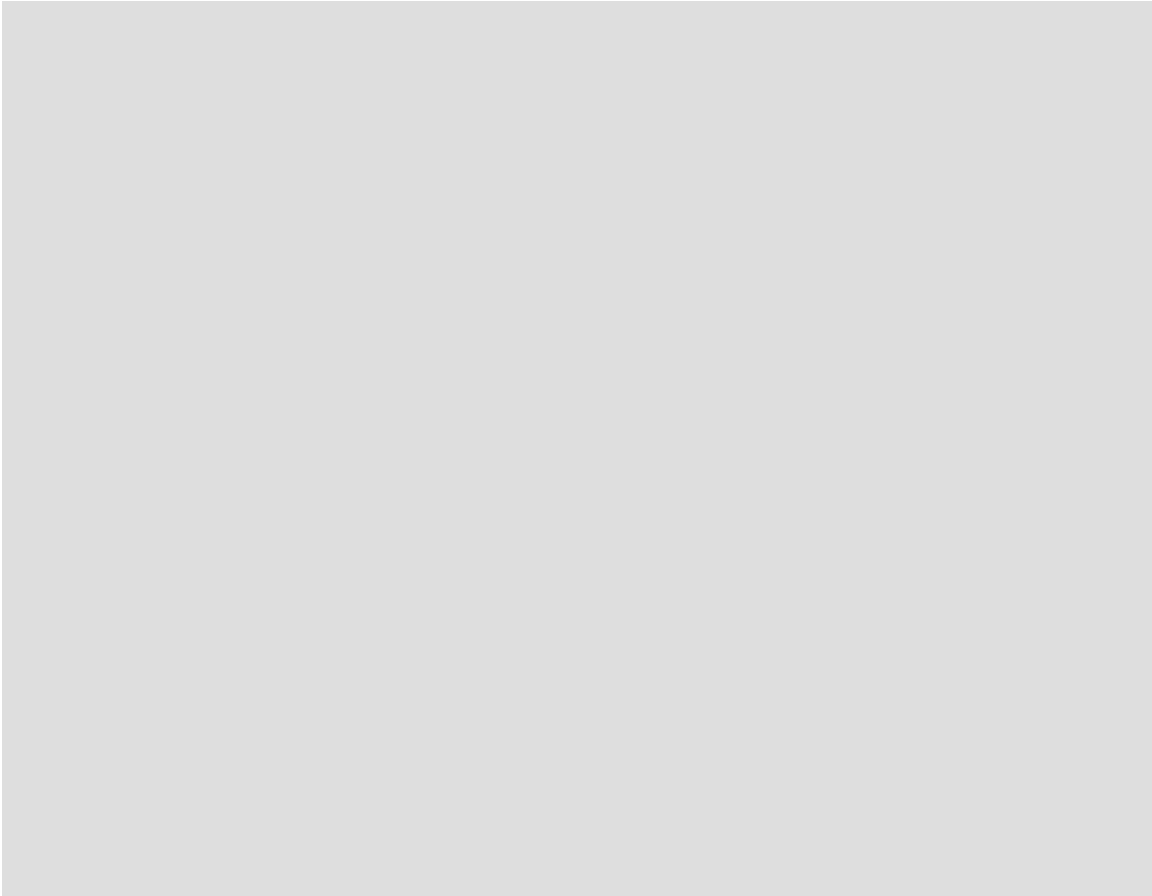
AI features shipped to 500 partners.

Q4

Scale Out

APAC regions live; global coverage.

Built in the Field



Built with customers.

60% faster; 50% fewer tickets

What We Offer

Insights

Live data
and AI
analytics.

Security

Zero-trust,
SOC 2
ready.

Speed

Sub-100ms
p99 latency.

APIs

300+ open
connectors.

Annual Strategy Review

Executive Leadership — January 2026

2025 Results at a Glance

127%

Revenue Target

\$3.8B total

94%

Retention Rate

48 NPS

Net Promoter

+12 pts YoY

What Drove Growth

Enterprise & Products

Enterprise up 42%

APAC beat forecast 31%

New products: 18%

Core & Channel

SMB churn down

Services attach: 34→51%

Partner channel: 28%

Strategic Priorities 2026

Q1

Foundation

Merge legacy systems; unified data layer.

Q2

AI Assist

AI workflows GA for all customers.

Q3

Expansion

Fortune 500 sales motion launched.

Q4

Open Beta

ISV program; target 50 partners.

Investment Areas

Build

Reliability
and dev
tooling.

Growth

Enterprise
sales and
channels.

Global

APAC and
EMEA
expansion.

Analytics

Customer
and usage
insights.

Component Gallery

All eight components in one deck

Icon Text Rows

Speed

Deploy in minutes.

Security

Safe by default.

Scale

Grows with you.

Stat Callout

99.9%

Uptime

< 50ms

Latency

p99 globally

500+

Integrations

Comparison Columns

Before

Manual; error-prone

Weekly reports

After

Automated; reliable

Real-time data

Timeline

Month 1

Kickoff

Setup and
team
training.

Month 3

Go Live

Full
production
rollout.

Month 6

Growth

Tuning and
expansion.

Image Half Bleed

Image with text side.



Card Grid

Feature A

Fast and
reliable.

Feature B

Secure
by
design.

Feature C

Scales
with you.

Feature D

Pay as
you go.



01

Where We Stand Today

What We Will Cover

- 01 Performance against targets
- 02 Customer and market signals
- 03 Operational health and risks
- 04 Roadmap for next quarter
- 05 Decisions we need today

3.2M

Active customers worldwide



"We buy outcomes, not tools."

- VP of Engineering, enterprise customer



Net Revenue Retention

Revenue kept from existing customers.



What if we trusted users?

What Changed This Quarter

Activation time dropped by half

Support tickets fell sharply

Expansion revenue outpaced new sales



hello.py

```
# The classic, end to end  
def main():  
    print("Hello, world!" )
```

```
main()
```

```
$ python hello.py  
Hello, world!
```

Three Steps to Onboard

01 Connect your data

Link a source in minutes with a guided wizard.

02 Pick a template

Start from a proven layout, then tailor it.

03 Share the result

Publish to your team and gather feedback fast.

Onboarding, Reworked

Before

Manual account setup

Three-day approval wait

Email-only support

After

One-click provisioning

Same-day approval

In-product live chat

Cloud or On-Prem

3 wks

Cloud time-to-launch

vs

6 mo

On-prem time-to-launch

Key Metrics This Quarter

98.7%

Uptime

1,240

New accounts

4.6/5

Support rating

\$2.1M

Net new revenue

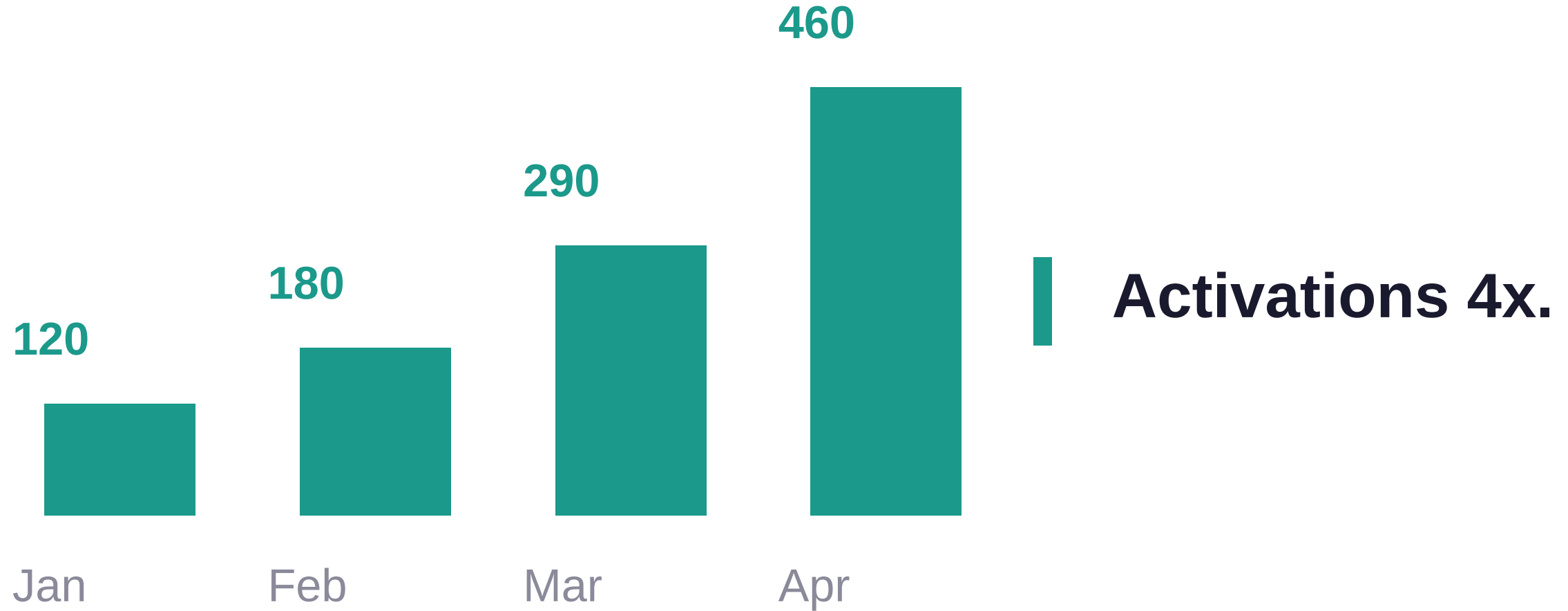
12 days

Time to value

31%

Expansion rate

Monthly Active Teams



Pricing Tiers

Plan	Seats	Price
Starter	Up to 5	\$0
Team	Up to 50	\$12/seat
Business	Up to 250	\$20/seat
Enterprise	Unlimited	Custom

Performance vs Last Year

\$8.4M

Revenue

+24%

2.1%

Monthly churn

-0.8 pts

63

Promoter score

+11

Onboarding Flow

01

Sign up

Create an
account

02

Connect

Link your
data

03

Configure

Pick your
metrics

04

Launch

Go live in
minutes

2026 Plan

Q1

Ship API

Beta launch

Q2

Mobile app

Integrations

Q3

Enterprise tier

SSO

Sales Pipeline

Visitors · 50,000

Signups · 8,000

Trials · 2,400

Customers · 600

Hierarchy of Needs

Vision

Strategy

Goals

Tactics

SWOT Analysis

Strengths

Strong brand, loyal users

Opportunities

New markets via partners

Weaknesses

Thin margins, small team

Threats

New entrants cut prices

Evaluation Matrix

	Platform A	Platform B	Platform C
Security	Strong	Medium	Strong
Workflow	Medium	Strong	Medium
Support	Strong	Strong	Medium

Job Search Funnel (2021)



Final folder shape

```
simple-eda-project/  
  ■■ src/  
    ■ ■■ simple_eda/  
      ■ ■■ __init__.py  
      ■ ■■ core.py  
    ■■ README.md  
    ■■ pyproject.toml  
    ■■ LICENSE
```

Agents in AI

How software learned to act, not just answer

What Makes Software an Agent

Perceive

Read its environment.

Reason

Plan a course of action.

Act

Use tools to change things.

Assistant vs Agent

Assistant

Answers one prompt

Waits for you

Has no tools

Agent

Pursues a goal

Acts on its own

Calls tools

Core Capabilities

Planning

Break goals into steps.

Tool Use

Call APIs and code.

Memory

Recall past context.

Reflection

Critique and retry.

Why Now

2022

LLMs Arrive

tool use emerges

10x

Faster

early reports

1M+

Developers

building agents

From Rules to Agents

1980s

Rules

Hand-coded.

2010s

ML

Learned patterns.

Today

Agents

Models that act.

Common Agent Patterns

ReAct

Reason,
then act.

Tool Use

Call
functions.

RAG

Retrieve
facts first.

Multi-Agent

Specialists
team up.

Multi-Agent Systems

Many narrow agents beat one generalist.

Each owns a clear role.

A coordinator routes the work.

They hand off context.

Promise vs Pitfalls

Promise

Automate real work

Scale expertise

Run all day

Pitfalls

Errors compound

Hard to predict

Needs guardrails

Takeaways

Software that acts, with human oversight

Claude Code



Claude Code

A terminal AI agent that writes and runs your code.

Chat vs Agent

Chat

You ferry code around

vs

Agent

It works in your repo

What It Can Do in Your Project

Explore

Read files, schemas, and notebooks.

Execute

Run scripts, queries, and tests.

Iterate

Check outputs and fix its own errors.



Terminal Use

```
$ cd sales-analysis && claude
```

```
> Profile sales.csv: nulls, dtypes,  
outliers. Save it to profile.md.
```

```
Reading sales.csv (1.2M rows)...
```

```
Running profiling script...
```

```
Wrote profile.md: 3 findings.
```

Everyday Workflows

EDA

Profile new
data in
minutes.

Cleaning

Fix types,
nulls,
duplicates.

Modeling

Fast first
baselines.

Reporting

Plots and
write-ups.

How It Works on a Task

01

Explore

Scans the data and code.

02

Plan

Proposes an approach.

03

Execute

Writes and runs the code.

04

Verify

Checks results, then iterates.

A Typical Day, Reworked

Before

Boilerplate pandas by hand

Stack Overflow detours

Stale notebooks

With Claude Code

Describe the outcome

Errors debugged in place

Committed with notes

Commands Worth Knowing

Command	What it does
<code>claude</code>	Start a session
<code>/init</code>	Write CLAUDE.md
<code>/compact</code>	Trim the context
<code>claude -p</code>	One-shot mode

Two Commands, Up Close

/init

Scans your project, writes a starter CLAUDE.md.

claude -p

One-shot: run a prompt, print the answer, exit.

Habits That Make It Shine

Keep a CLAUDE.md with data caveats.

Ask for a plan before a big analysis.

Have it verify outputs, not just run.

Commit small, reviewable steps.



01

One Claude, Many Surfaces

Same Engine, Everywhere

One engine: terminal, IDE, desktop, web.

Your CLAUDE.md and MCP servers come too.

Start on the laptop, finish on your phone.

Three Ways to Run It

	Terminal	Web	Desktop
Runs on	Your machine	The cloud	Your machine
Setup	Install CLI	None	Install app

On the Internet

Cloud-run

Runs at `claude.ai/code` on Anthropic's cloud.

Isolated

Each task gets its own sandbox VM.

Persistent

Keeps running after you close the tab.

Claude Desktop

A standalone app, no terminal needed.

Review diffs, run sessions side by side.

Schedule repeats and start cloud runs.



02

Drive It From Your Phone

Connect GitHub

01 Sign in

Open `claude.ai/code` and log in.

02 Link GitHub

Grant access to the repos you choose.

03 Pick a repo

Claude clones it into a cloud sandbox.

Start a Task Anywhere

01 Describe it

Type the task in the Claude mobile app.

02 Let it run

Work continues in the background.

03 Review

Approve the branch Claude pushes.

Remote Control

Step away and work from your phone.

Continue a live session from any browser.



Move Between Surfaces

```
$ claude --cloud
```

Runs this task **in** the cloud.

> Pick it up later, anywhere:

```
$ claude --teleport
```

One session follows you across
laptop, phone, **and** browser.



03

MCP: Plug In Your Tools



MCP

An open standard linking AI tools to your data.

What MCP Unlocks

Read design docs from Google Drive.

Update Jira, pull data from Slack.

Query your own database and APIs.

Without MCP vs With MCP

Without MCP

Paste data in by hand

Screenshot dashboards

Describe your schema

With MCP

Claude queries the DB

Reads the ticket itself

Always current data

Servers You Can (normally) Connect to

Server	What Claude can do
Drive	Read design docs
Jira	Update tickets
Slack	Pull conversations
Postgres	Query your data



04

Customize Claude



CLAUDE.md

A project file Claude reads every session.

What Goes in CLAUDE.md

Coding standards and libraries.

Architecture and key commands.

Review checklists and data caveats.

Example CLAUDE.md

sales-analysis project

- Use pandas, not polars.
- Raw data lives in data/.
- Never commit the CSVs.
- Tests: pytest -q
- Plots: matplotlib only.

CLAUDE.md Pitfalls

Too long crowds the context.

Stale rules Claude still obeys.

Vague advice it cannot act on.

Secrets do not belong here.



Skills

Reusable instruction folders, loaded on demand.

CLAUDE.md vs Skills.md

CLAUDE.md

Always in context

Project-wide rules

One file at the root

Skills.md

Loaded on demand

A packaged workflow

Shared across projects



05

More From Claude



Cowork

Claude Code in Claude Desktop, for non-code work.

What Cowork Does

Reads and edits files you choose.

Works across Slack and Google Drive.

Finishes multi-step tasks, with citations.



Artifacts

Shareable code, docs and apps built beside chat.

Memory > Chat Search

Memory

Learns your context

Persists across chats

Separate per project

Chat search

Looks up old chats

Only when you ask

One lookup at a time

| Build it from our phones.

slidekit First Light

Phase 5 demo: lint-clean, zero screenshots

How slidekit Works

Author IR YAML once

Layout engine solves EMU rects

Linters validate before python-pptx emits

Round-trip test verifies — zero screenshots

Calibration Results

0.076%

Max Width Error

vs LibreOffice render

2%

Error Threshold

hard gate PASS

122

Tests Passing

phases 1 through 5

Design Principles

Deterministic

Same input always produces same output.

No screenshots

Lint errors cite path and exact fix.

Agent-ready

Build and fix loop needs no images.

Before and After

Ad-hoc pptx code

Manual EMU math

Visual QA required

No lint feedback

Agent loops on images

slidekit

Declarative IR YAML

Round-trip verified

Lint errors with fix

Agent loops on text

Build Your First Library

The path to PyPI

01

Function

pandas
code

02

Package

one folder

03

Build

make a
wheel

04

Publish

pip install

Today's goals

01 Package

Create a real, importable Python package.

02 Publish

Upload it to TestPyPI, then PyPI.

03 Install

`pip install` your own package.

| Build tiny. Publish safely.



What we're building

```
import pandas as pd  
import simple_eda as eda
```

```
df = pd.read_csv("data.csv")  
eda.summarize(df)  
eda.missing(df)
```

What we build now — and skip

Build now

One dependency: pandas

Plain Python functions

A clear README

Skip for now

CI/CD

Testing

Many dependencies



01

What is a Python library?



Library

Reusable code you install once and import anywhere.

Script vs library

Script

`python analyze.py`

Less convenient

Limited scope

Library

`import simple_eda`

More convenient

Broad scope

A tiny library is still real

No web app required.

No dashboard required.

Just useful, installable, importable.

The five bare minimums

A unique package name

A project folder

An `__init__.py` file

Your functions

Metadata — `pyproject.toml`



02

Design the smallest API



API

The functions and classes users call.



MVP

The smallest version that is genuinely useful.



the simplest API

```
import simple_eda as eda  
eda.summarize(df)
```

Pick boring function names

Prefer (boring)

summarize

missing

numeric_columns

Avoid (clever)

hyper_profile_engine

magic_diagnose

mega_summary

MVP Input: Pandas dataframe only

Support pandas DataFrames first.

Not Polars, Spark, Dask, or Arrow yet.

Fewer choices means fewer bugs.

MVP Output: Plain objects only

Dictionaries & lists

Simple built-in containers.

Ints & strings

Plain scalar values.

pandas Series

No custom classes to learn.

The 0.1.0 mindset

First working version, not perfect.

Installable and understandable.

Momentum, not completeness.



03

Create the package folder

Final folder shape

```
simple-eda-project/  
  ■■ src/  
    ■ ■■ simple_eda/  
      ■ ■■ __init__.py  
      ■ ■■ core.py  
    ■■ README.md  
    ■■ pyproject.toml  
    ■■ LICENSE
```

Install name vs import name

Install name

simple-eda-you

hyphens

used by pip & PyPI

Import name

simple_eda

underscores

used in Python code

Why the names differ

import needs a valid Python name.

Python names cannot contain hyphens.

So the import name uses underscores.

PyPI treats - and _ as the same name.



`__init__.py`

Marks a folder as an importable package.

Why the src/ layout?

Blocks accidental root imports.

Matches modern Python projects.

Makes packaging behavior clear.

Common folder mistakes

pyproject.toml in the wrong folder.

Forgetting `__init__.py`.

Folder named `simple-eda`, not `simple_eda`.

Installing from the wrong directory.

What each mistake causes

Mistake	What happens
No <code>__init__.py</code>	import cannot find it
Hyphen in folder	invalid module name
pyproject misplaced	build finds nothing
Wrong directory	ModuleNotFoundError



04

Write the EDA functions



```
def summarize(df):  
    return {  
        "rows": len(df),  
        "columns": len(df.columns),  
        "names": list(df.columns),  
    }
```

The core functions

summarize — shape, columns, types

missing — nulls per column

numeric_columns / categorical_columns



missing()

```
def missing(df):  
    counts = df.isna().sum()  
    return counts.sort_values(  
        ascending=False)
```

```
# missing(df) ->  
# age      2  
# name     1
```

numeric & categorical columns

```
def numeric_columns(df):  
    sel = df.select_dtypes(  
        include="number")  
    return list(sel.columns)
```

```
def categorical_columns(df):  
    sel = df.select_dtypes(  
        include=["object", "category"])  
    return list(sel.columns)
```



Try it on a tiny DataFrame

```
import simple_eda as eda
```

```
df = pd.DataFrame({  
    "name": ["Ana", "Bo", None],  
    "age": [25, None, 31],  
})  
print(eda.summarize(df))  
print(eda.missing(df))
```

Function do's and don'ts

Do

Return data

Use plain dicts / Series

Keep signatures small

One job per function

Don't

Change df in place

Print instead of return

Add many optional args

One function for everything

One job vs everything

One job each

Small and focused

Easy to test

Easy to reuse

One big function

Hard to follow

Hard to test

Breaks more often

Signature

A function name plus the parameters it takes.



05

Add pyproject.toml

What pyproject.toml does

Names your package and version.

Lists dependencies.

Tells build tools how to build.



Build tools

Tools that turn source into installable packages.

Minimal pyproject.toml

```
[build-system]
```

```
requires = ["setuptools>=61"]
```

```
build-backend = "setuptools.build_meta"
```

```
[project]
```

```
name = "simple-eda-yourname"
```

```
version = "0.1.0"
```

```
description = "Tiny EDA helper"
```

```
dependencies = ["pandas>=1.5"]
```



TOML

The simple key = value format pyproject uses.

Version numbers

Version	Means
0.1.0	First working version
0.1.1	A small fix
0.2.0	A new feature
1.0.0	Stable and public



MAJOR.MINOR.PATCH

Each of the three numbers is a non-negative integer.

README: install & use

simple-eda-yourname

Tiny EDA helper for pandas.

Install

```
pip install simple-eda-yourname
```

Use

```
import simple_eda as eda  
print(eda.summarize(df))
```

License basics

Says what others may do with your code.

MIT is common for small projects.

Use an official template.

Never invent your own license text.



06

Install locally with pip

Install in editable mode

```
# run from the folder with pyproject.toml  
python -m pip install -e .
```

What -e (editable) means

Links pip to your source folder.

Code edits apply instantly.

No reinstall while developing.

Smoke-test the import

```
>>> import simple_eda as eda  
>>> eda.__all__  
['summarize', 'missing', ...]
```

When the import fails

Symptom

ModuleNotFoundError

Wrong folder

Wrong environment

Fix

Run `pip install -e .`

`cd` to the project root

Use `python -m pip`



07

Build a real package file

Build the artifacts

```
python -m pip install build
```

```
python -m build
```

The dist/ folder

dist/

■■■ simple_eda_yourname-0.1.0.tar.gz

■■■ simple_eda_yourname-0.1.0-...whl

Wheel vs source distribution

Source (.tar.gz)

A source archive

Used as a fallback

Wheel (.whl)

Prebuilt, fast install

What most users get

Keep build output out of git

dist/, build/, *.egg-info are generated.

Add them to .gitignore.

Bump the version before rebuilding.



08

Publish safely

Why TestPyPI first?

A practice package index.

Rehearse upload and install commands.

Mistakes are cheap here.

The upload flow

01 Check

```
python -m twine check dist/*
```

02 TestPyPI

```
twine upload --repository testpypi dist/*
```

03 PyPI

```
twine upload dist/* — only after TestPyPI works.
```

Install from TestPyPI

```
# install from TestPyPI
```

```
pip install \
  -i https://test.pypi.org/simple/ \
  --extra-index-url \
    https://pypi.org/simple/ \
  simple-eda-yourname
```

Publishing safety

Use API tokens, not passwords.

Never commit tokens or secrets.

Fewer deps, less attack surface.

Beware typosquatting.



09

Use, improve & share

What to add — and avoid

Avoid early

Full HTML reports

Plugin systems

Every dataframe backend

AI-generated analysis

Good next steps

`quick_report()`

`top_categories()`

`column_summary()`

`duplicate_rows()`



Add a tiny test

```
def test_numeric_columns():  
    df = pd.DataFrame(  
        {"name": ["A"], "age": [10]})  
    cols = eda.numeric_columns(df)  
    assert cols == ["age"]
```

Final checklist

Package files all exist.

`pip install -e .` works.

`python -m build` works.

`twine check dist/*` passes.

README has install + usage.

0.1.0



Ship your first version

Four functions, importable, published.

From Vague to Precise

Immersion Week SF · Day 1 · Session 1.1

35



Minutes

You leave with one precise problem statement

What This Session Covers

- 01** Problem statements versus solution statements
- 02** Making Claude interrogate a vague ask
- 03** The three-gate scoping test
- 04** Standing up your Claude Project

| What are you building?

Two Ways to Answer That

Vague

An AI thing for resumes

vs

Precise

Flags gaps in 200 files

How We Get There

01

Separate

Problem
apart from
solution

02

Interrogate

Let Claude
attack the
ask

03

Scope

Run the
three-gate
test

04

Commit

Write it into
the Project



01

Problem, Not Solution



Problem statement

A gap between what happens and what should.



Solution statement

A thing you plan to build, assuming the diagnosis.

How to Tell Them Apart

Problem

Describes a gap

Admits several fixes

Names who suffers

Solution

Describes a build

Assumes one fix

Names a tool

What They Said, What They Meant

Said (solution)

Meant (problem)

Build a RAG chatbot

Staff cannot find answers

Fine-tune a model

Output misses our tone

Make a dashboard

Nobody sees the numbers

Use an agent

Review takes four hours

Why Solution-First Hurts

You cannot tell if it worked

The tool choice is already locked

Scope creeps with no boundary

Feedback arrives too late to use

Three Tells You Wrote a Solution

A tool is named

RAG, agent, fine-tune, dashboard

Only one fix fits

No alternative is even thinkable

No one is in it

Nobody is described as stuck



"Names a tool? Not a problem yet."

- The fastest test in the room

Interrogation

Making Claude attack a vague ask until it is precise.

What Makes Claude Interrogate

Give it the adversarial role

Forbid it from proposing solutions

Demand one question at a time

Make it refuse to move on early



The Interrogation Prompt

You are a skeptical product lead.
My idea: "an AI thing for resumes"

Rules:

- Ask ONE question at a time.
- Never propose a solution.
- Reject vague answers.
- Stop when it **is** precise.

The Interrogation Loop

01 Answer plainly

No hedging, no tool names

02 Let it push

Expect the follow-up to hurt

03 Restate

Ask Claude to write it back

Before and After the Interrogation

Before

An AI thing for resumes

Who for? Unclear

Done when? Unclear

After

Recruiters miss skill gaps

For 200 intern applicants

Flagged in one pass

Signs It Is Now Precise

A stranger can restate it

You know who is stuck

You can name a wrong answer

Two people would agree it is solved



02

The Scoping Test

Scoping test

Three gates a problem passes before you commit.

The Three Gates

Measurable outcome

You can say whether it worked

Data available

You have it, today, in hand

Three days

Finishable by Friday noon



Gate 1 · Measurable

State up front what would prove it worked.

Vague Metric, Measurable Metric

Vague

Better resumes

Faster reviews

Higher quality

More useful

Measurable

Flags 8 of 10 known gaps

Under 30 seconds per file

Two reviewers agree it passes

Used unprompted twice



Gate 2 · Data

It exists, you may use it, you have it today.

Where Data Plans Die

Access pending

Someone still has to approve it

Wrong shape

Exists, but not in usable form

Too sensitive

Real records you cannot touch



Gate 3 · Three days

The whole path working, not one part polished.

What Three Days Buys

One path

End to end and rough

vs

One part

Polished but alone

Three Candidate Ideas, Three Gates

	Resumes	Policy	Grader
Measurable	Yes	Yes	Yes
Data now	Yes	No	Yes
3 days	No	Yes	No

The opposite case

Making Claude argue your problem is the wrong one.



Make Claude Argue Against You

Our problem statement:

"<paste it>"

Argue this **is** the WRONG problem
for a three-day build.

Three strongest objections,
hardest first. No alternatives.

Running the Opposition

01 Read all three

Do not answer while reading

02 Rank them

Which would actually sink you

03 Fix or accept

Narrow it, or note the risk

When the Objection Wins

Narrow the users, not the ambition

Shrink the data, keep the path

Stub the hard step for now

Change problems only if a gate fails

Cut Scope, Not Ambition

One user type, not four

One input format, not any

One clean path, no edge cases



03

The Team's Shared Brain



Claude Project

A shared workspace holding instructions and context.

Chat Versus Project

A chat

Context dies at the tab

One person holds it

Re-explained every time

A Project

Context persists

The whole team shares it

Explained once

What a Project Holds

Instructions

Standing rules for every chat

Context

Uploaded docs and data

One place

Everyone works from it

Stand One Up Now

01 Create

Name it for the problem

02 Instruct

Paste the problem statement

03 Share

Add every teammate today

Project Instructions Template

Problem

<the precise statement>

Success looks like

<the measurable outcome>

Out of scope

<what we cut, and why>

What to Upload

Sample data, even if faked

The interrogation transcript

The objections you accepted

What Not to Upload

Real personal records

Anything under an NDA

Credentials or API keys

Team Conventions Worth Setting

One source

Project over
chat

Log it

Decisions
land in
writing

Shared

Everyone
reads it first

Update

Change it
when scope
does

Project Pitfalls

Instructions written once, never updated

Uploads nobody has actually read

One member still working in chat

Done With This Session When

The statement passes three gates

The strongest objection is answered

The Project exists and is shared

What You Leave With

1

Precise problem

3

Gates passed

1

Shared Project

Next: Mapping the Workflow

Break the problem into steps

Draw it as a Claude Artifact

Name what each step needs

| Can a stranger restate it?

Interface Aesthetics

Ngo, Teo & Byrne · Information Sciences (2003)



Is beauty computable?



Interface aesthetics

Layout quality, written as numbers.

14

Aesthetic measures

Closed-form functions of element bounding boxes

The core claim

Layout quality is not purely subjective.

Balance and spacing become numbers.

Each number is a formula over boxes.

Measured, optimised, checked vs people.



01

The fourteen measures

The measures (1–5)

Balance — even visual weight

Equilibrium — mass near the centre

Symmetry — mirrored about the axes

Sequence — a clear reading flow

Cohesion — aspect ratios agree



Bounding box

Every measure reads position and size.

The measures (6–10)

Unity — parts read as one

Proportion — pleasing ratios

Simplicity — few alignment points

Density — comfortable coverage

Regularity — elements snap to a grid

The measures (11–14)

Economy — few sizes, reused

Homogeneity — evenly distributed

Rhythm — regular, repeating change

Order & complexity — structure wins

regularity — Ngo (2003)

Ngo regularity (alignment term)

$$\text{regularity} = 1 - (n_{\text{vap}} + n_{\text{hap}}) / (2n)$$

a tidy 5-bullet column:

$n = 5$, $n_{\text{vap}} = 1$, $n_{\text{hap}} = 5$

$$\text{regularity} = 1 - (1 + 5) / 10 \quad \# 0.40$$

slidekit: left edges only

$$\text{regularity} = 1 - 1 / 5 \quad \# 0.80$$

Same clean column, two scores

0.40

Ngo x + y regularity

vs

0.80

Left-edge alignment

Why slidekit fits this paper

Every element already has an exact rect.

Measures are arithmetic over geometry.

Deterministic — same deck, same score.

Zero rendering, zero vision.



02

From paper to slidekit

What slidekit adopted

Declined

Regularity

Rhythm

Proportion

Economy

Spacing-regularity

Adopted

Balance

Density

Adoption decisions

Measure	Verdict	Reason
Balance	Adopted	Axis moment
Density	Adopted	Coverage
Regularity	Declined	Halves cols
Proportion	Declined	Ratio noise

A measured-aesthetics lineage

1933

Birkhoff

A beauty
ratio

2003

Ngo

Fourteen
measures

2004

Lok

Visual
balance

2026

Zhang

Moment
model

Honest scope

The journal paper is paywalled.

Public copies reproduce only alignment.

Constants are not copied blindly.

No universal beauty equation is claimed.

From measure to code

01 Read geometry

Take each element's rect.

02 Define a formula

Write an open, closed-form score.

03 Verify & ship

Check the source, then report.



"Make design quality auditable."

- Ngo, Teo & Byrne (2003)